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1. Executive Summary

This report maps every practical way to gather job postings programmatically in Germany for a single, urgent use case: Asha, a freshly qualified Fachinformatiker für Systemintegration (FISI), needs his first junior IT job fast in the weakest junior hiring market in years. The research behind this document was performed on 11-12 October 2026 with six parallel deep-research tracks and live, first-party verification of every endpoint that matters. Every claim of the form this API works in this report was tested with a real HTTP request during the research window, and every dead end was verified as dead so that no time is wasted building on deprecated infrastructure.

The headline finding is positive: a complete, fast, legally safe job-watching pipeline can be built today using exclusively green-tier sources. The single most valuable asset is the Bundesagentur für Arbeit (BA) Jobsuche API, a free official service that requires no registration and currently returns 238 Fachinformatiker Systemintegration postings within 50 km of Erlangen and roughly 6,755 Systemintegration postings Germany-wide. Around this backbone we assemble keyed aggregator APIs (Adzuna, Jooble), a no-key German board API (Arbeitnow), two verified low-volume portal channels (XING server-rendered search results and the LinkedIn guest endpoint), direct employer ATS feeds at DATEV, Schaeffler, adidas, GfK, PUMA, Siemens Healthineers, Hetzner and more, plus a remote-work RSS suite. Portals that actively block automation (StepStone, Indeed, meinestadt) are covered by their native email alerts instead of scraping.

238	6,755	8x	30+
FISI postings within 50 km of Erlangen (BA API, live count)	Systemintegration postings Germany-wide (BA API)	more interviews when applying within 72 h of posting	programmatic sources mapped and risk-rated

Because Asha commutes by train from Erlangen, the report includes a train-reachability filter derived from the official VGN GTFS open dataset: a verified whitelist of 29 municipalities reachable with zero transfers in under 60 minutes, plus explicit exclusions such as Herzogenaurach, which despite hosting PUMA and adidas has no rail station at all. The final chapters assemble these components into a concrete CSV-tracker pipeline on Python plus cron, a portal-native alerting checklist, a legal risk matrix, and a 14-day action plan. The overall design principle is simple: automate finding and filtering completely, and reserve human effort for applying quickly and well.

Pipeline layer	Primary sources	Why it wins
Backbone	BA Jobsuche API v6, EURES API, Arbeitnow	Free, no key or public static key, hourly freshness, covers all of Germany incl. the BA-Jobbörse long tail
Aggregators	Adzuna DE, Jooble, Careerjet v4	One query reaches many boards; server-side keyword and geo filtering; salary fields
Portal channels	XING SSR extraction, LinkedIn guest endpoint	Real-time portal coverage without login; used at low personal-use volume (yellow tier)
ATS-direct	Personio /xml, SuccessFactors feed.xml, Workday CXS, SmartRecruiters	Zero competition: watch target employers before postings reach portals
Remote suite	WeWorkRemotely, Himalayas, Remotive, RemoteOK	Fully remote fallback positions across Germany and Europe
Native alerts	StepStone Jobagent, Indeed/meinestadt emails, XING Job-Alerts	Covers anti-bot portals legally through their own alert products

Table 1-1: The Speed Stack at a glance. Each layer is detailed in its own chapter.

2. Target Profile and Search Strategy

Asha's profile defines the search surface. He completed an Ausbildung as Fachinformatiker für Systemintegration, which in German postings appears under a predictable cluster of titles: FISI roles, IT-Systemelektroniker, Systembetreuer, junior IT-Administrator, First/Second-Level Support and MSP service technician positions. The scope agreed for this search is FISI plus adjacent junior IT, which is deliberately wider than strict title matching because many employers title these jobs as IT-Support or Systemadministrator while the actual work is classic Systemintegration: client and server rollout, Windows domains, networking, ticket handling and customer-site service.

Market reality: why speed is the whole game

The 2025/2026 junior IT market is structurally hostile to newcomers. Bitkom counted roughly 149,000 open IT positions in 2023 but only about 76,000 in 2024, and the senior overhang from tech layoffs has crowded entry-level pipelines. The BA Fachkräfteengpassanalyse keeps technical IT professions in the shortage or observation zone, but the shortage is concentrated in experienced profiles; juniors face the bottleneck. The practical consequence is that Asha needs both volume and speed: 40 to 80 high-quality applications, and a repeatable process that surfaces each new relevant posting within hours rather than days.

Timing evidence justifies the whole engineering effort. The widely cited TalentWorks analysis of roughly 1,600 applications found that applying within the first 72 hours of a posting yields up to eight times more

interviews than applying late. More recent aggregate data from careery (2026) puts the median employer first-response at 6.7 days, and huntr data shows an 11.2 percent interview rate for direct company applications. Together these numbers define the pipeline requirement: detect new postings within 24 hours, apply within 72, and never hear about a good posting a week late from a daily digest alone.

2.800-3.900 €	6.7 d	11.2 %
typical FISI entry salary range, brutto/month (ausbildung.de)	median first response across applications (careery 2026)	interview rate for direct company applications (huntr)

Certifications that shorten the search

Two certifications give the best effort-to-signal ratio for FISI applicants. Microsoft AZ-900 can be earned in one or two weekends and appears in a large share of junior infrastructure postings as a baseline cloud credential. ITIL 4 Foundation matches the vocabulary of servicedesk-heavy roles and is frequently used as an HR filter keyword by MSPs. CCNA is the stretch goal for the Netzwerkadministrator track and pays off specifically in Systemhaus environments. LPIC-1 is respectable but carries less recognition in German HR filtering than the Microsoft line.

Keyword engineering

API queries are only as good as their keyword sets. The table below is the copy-paste-ready search vocabulary for the whole pipeline. It separates core German role names, additional German variants that appear in real postings, English-language variants increasingly used in German IT ads, employer-type terms for discovering MSPs and Systemhäuser, and the exclusion list that keeps senior and unrelated roles out of the tracker. Query syntax differs per source and is noted per row; the BA API takes one string per query, so the pipeline loops over the keyword list rather than trying to build one OR query.

Category	Terms	Usage notes
DE core	Fachinformatiker Systemintegration; Systemintegration; IT-Systemelektroniker; Systembetreuer; Systemadministrator; IT-Administrator; IT-Support; First-Level-Support; Second-Level-Support; IT-Servicedesk; IT-Techniker; Client Management; Netzwerkadministrator; Serveradministrator; Cloud-Administrator; Junior DevOps; Anwendungsbetreuer; IT-Service	BA API: one was= string per query, loop over all; Adzuna what=; LinkedIn keywords=
DE extra	IT-Helpdesk; Helpdesk-Mitarbeiter; IT-Betreuung; IT-Mitarbeiter; IT-Infrastruktur; Infrastruktur-Administrator; Systemadministration; IT-Systemtechniker; IT-Supporter; IT-Anwenderbetreuung; Junior Systemadministrator	Second wave of BA queries; same loop
EN variants	IT Support Specialist; System Administrator; Junior System Engineer; IT Operations; Service Desk; Helpdesk Technician	Common in international and corporate postings; Adzuna and LinkedIn respond well
Employer types	Systemhaus; IT-Dienstleister; Managed Service Provider; MSP; IT-Serviceprovider; Rechenzentrum	Used for building the employer watchlist (Chapter 7), not for title search
Exclusions	Senior; Principal; Lead; Head of; Teamleitung; Architekt; 5 Jahre Berufserfahrung; mehrjährige Berufserfahrung; Werkstudent; Praktikum; Ausbildung; Azubi; Vertrieb; Sales; Student; Freelance; freiberuflich	Negative filter in the tracker scorer; Ausbildung and Praktikum are excluded because Asha has completed his Ausbildung

Table 2-1: The keyword set for the pipeline. Semicolons separate individual terms.

3. Provider Coverage Matrix

This chapter is the master map. More than 30 German and Germany-covering providers were investigated; the table below condenses each to its access method, authentication requirement, freshness, and legal risk tier. Green means a source is explicitly offered for consumption (official API, public feed, RSS) and can be polled on a schedule. Yellow means a technically accessible channel that the provider's terms do not bless, safe only at low personal-use volume. Red means actively defended infrastructure where automated access is both contractually banned and technically punished; red sources are handled through native alerts instead. The striking result of the research: the green tier alone already covers the vast majority of reachable FISI postings in Asha's region, so the pipeline never needs to touch a red source.

Provider	Method	Auth	Freshness	Risk	Verdict
Bundesagentur für Arbeit	REST API v6 (jobsuche-service)	static public key	realtime	green	Backbone. 238 FISI hits near Erlangen, detail endpoint per job

Provider	Method	Auth	Freshness	Risk	Verdict
EURES (EU portal)	JSON search API	none	realtime	green	8,499 FISI records; BA mirror plus EU extras
Arbeitnow	JSON API, no key	none	hourly	green	German board; client-side filtering required
Adzuna	REST API /jobs/de	free app key	realtime	green	Best queryable API; 250 calls/day free tier
Jooble	POST API	free key	realtime	green	Simple keywords+location search
Careerjet	Partner API v4	partner key	realtime	green	Legacy keyless API closed to new users
WhatJobs	Publisher API	publisher key	realtime	green	Affiliate program required
XING Jobs	SSR page JSON (APOLLO_STATE)	none (logged out)	realtime	yellow	16+ jobs per page incl. salary, one-click flag; robots.txt disallows /jobs/search so keep volume tiny
LinkedIn Jobs	Guest API seeMoreJobPostings	none (logged out)	realtime	yellow	Works unauthenticated; throttle hard, never with account cookies
StepStone	none; native Jobagent email	n/a	daily digest	red	Akamai bot manager verified; do not scrape
Indeed DE	none; native alerts	n/a	daily digest	red	API off since 2024, RSS dead (403), login wall
meinestadt.de	none; native alerts	n/a	daily	red	Akamai 403 on all endpoints
Jobware	none	n/a	n/a	red	403/connection kill
Monster.de	none	n/a	n/a	red	DataDome; inventory shrinking post-bankruptcy
glassdoor.de	none	n/a	n/a	red	Cloudflare; merged into Indeed 07/2026
GermanTechJobs	none	n/a	n/a	red	Old API 302-redirects to a paid jobcopilot signup
kimeta, JOBworld, jobs.de, jobtensor, Get-in-IT, heise/t3n/Golem jobs	none verified	n/a	n/a	red	Bot walls, 404/503; no individual-access API
Personio-hosted careers	GET /xml per company	none	realtime	green	Verified at NürnbergMesse, ITONICS, Nerdware

Provider	Method	Auth	Freshness	Risk	Verdict
SuccessFactors careers	GET feed.xml per company	none	realtime	green	Verified at Schaeffler (5.6 MB) and adidas (5.9 MB)
Workday CXS	POST .../wday/cxs/...	none	realtime	green	Verified at DATEV
SmartRecruiters	public API per company	none	realtime	green	Verified at GfK via NielsenIQ slug
PUMA careers	JSON search endpoint	none	realtime	green	594 jobs in one response
Siemens Healthineers	career sitemaps	none	weekly-ish	green	Phenom API blocked; sitemap fallback works
Hetzner	career sitemap	none	realtime	green	142 URLs incl. live FISI pages
WeWorkRemotely	RSS per category	none	realtime	green	DevOps/sysadmin feed fits FISI
Himalayas	RSS	none	realtime	green	Simple, reliable
Remotive	JSON API	none (attribution)	24 h delay	green	Free tier requires attribution; paid tier exists
RemoteOK	JSON API	none	realtime	green	Tags include junior and sysadmin
VGN GTFS (transit)	GTFS zip download	none	quarterly	green	Powers the train-reachability filter (Chapter 9)

Table 3-1: Master coverage matrix. Live verification status as of 11-12 October 2026.

Three structural observations emerge from the matrix. First, Germany's public-sector data infrastructure is dramatically more open than its commercial portal sector: the two most powerful sources (BA and EURES) are government services. Second, the aggregator tier is thin but sufficient; Adzuna alone aggregates a meaningful share of the market and offers proper server-side filtering, which the no-key German APIs lack. Third, the ATS layer is the hidden gem: major Franconian employers publish machine-readable feeds that most competitors never watch, which is exactly where an applicant seeking speed gains an edge.

4. Tier 1 - Official Backbone APIs

4.1 Bundesagentur für Arbeit Jobsuche API

The BA jobsuche service is the most important single discovery in this research. Much of the internet still documents the old endpoint rest.arbeitsagentur.de/jobads/jobs with the key jobsearch-beta; that combination is dead and returns HTTP 403 on every variation tested. The current production API, confirmed against the BA's own web client configuration, is versioned under jobboerse/jobsuche-service and accepts a static public client key, jobboerse-jobsuche. The official OpenAPI specification at jobsuche.api.bund.dev states that this key is public, that the API is free and that no registration is required, which makes this the lowest-friction high-volume job data source in Germany.

BA search - verified live 2026-10-12, HTTP 200

```
curl -sS -H "X-API-Key: jobboerse-jobsuche" \
  "https://rest.arbeitsagentur.de/jobboerse/jobsuche-service/pc/v6/jobs
?was=Fachinformatiker%20Systemintegration&wo=Erlangen&umkreis=50
&page=1&size=100&veroeffentlichtseit=7&angebotsart=1"

# detail page for one posting (base64 of referenznummer):
curl -sS -H "X-API-Key: jobboerse-jobsuche" \
  "https://rest.arbeitsagentur.de/jobboerse/jobsuche-service/pc/v4/
jobdetails/MTAwMDEtMTAwMzgzMjk5My1T"
```

Search parameters verified against the live service: was (keyword), wo (location), umkreis (radius km), page (1-based), size (max 100), veroeffentlichtseit (days 0-100, the freshness filter), angebotsart (1 = Arbeit, 4 = Ausbildung, 34 = Praktikum), arbeitszeit (vz/tz/snw/ho/mj), befristung, zeitarbeit, pav, behinderung, arbeitgeber (exact, case-sensitive) and berufsfeld. Old parameter names from the jobads era such as publishedSince are silently ignored. The search response contains the list under ergebnisliste plus maxErgebnisse, page, size, woOutput (geocoding) and facetten. Each item carries referenznummer, stellenangebotsTitel, firma, hauptberuf, alleBerufe, stellenlokationen with ort/plz/region and coordinates, entfernung (distance in km), datumErsteVeroeffentlichung, aenderungsdatum, eintrittszeitraum, gehaltsspanneVon/Bis, homeofficemoeglich, quereinstiegGeeignet and externeURL. Note that hashId and externBoersenId from the old API no longer exist: the detail key is base64(referenznummer), and private-board referrals are detected via externeURL presence.

Query (live tested)	maxErgebnisse
Fachinformatiker Systemintegration, Erlangen, 50 km	238
Fachinformatiker Systemintegration, Nürnberg, 50 km	237
Systemintegration, Erlangen, 50 km	305 (225 with angebotsart=1)
Systemintegration, Erlangen, 50 km, veroeffentlichtseit=7	56 (1 within 24 h)
IT-Systemelektroniker, Erlangen 50 km / Nürnberg 50 km	35 / 32
IT-Support, Nürnberg, 50 km	316
IT-Administrator, Erlangen, 50 km	96
berufsfeld=Informatik, Erlangen, 50 km	478
Systemintegration, Germany-wide	6,755
Fachinformatiker Systemintegration, Germany-wide	5,190

Table 4-1: Live result counts from the BA v6 API (2026-10-11/12). Erlangen and Nürnberg 50 km circles overlap almost completely, so one Nürnberg query covers the metro.

Operational notes matter for a cron pipeline. Twelve rapid consecutive requests all returned 200 during testing and every response carries an X-API-BLOCKED header (observed value 0), so an explicit blocking mechanism exists even though no numeric limits are documented. The BA terms of use prohibit automated access contrary to the intended purpose and bulk redistribution; a personal tracker polling a keyword loop every 30 to 60 minutes with caching by referenznummer is comfortably within intended use. Community libraries worth knowing: bundesAPI/jobsuche-api (161 stars, maintained), wunderfrucht/jobsuche (Rust, documents quirks such as sporadic 403-as-rate-limit and no sort parameter), long39ng/jobsuche (R), kemdev/jobsuche-api-js (TypeScript) and an n8n node, hansdoebel/n8n-nodes-jobsuche. There is no live Python package on PyPI, which is why Chapter 10 ships a minimal requests-based fetcher.

4.2 EURES search API

The EU EURES portal exposes an unauthenticated JSON search API that mirrors BA data and adds EU-wide extraditional listings. A POST to europa.eu/eures/api/jv-searchengine/public/jv-search/search with a keywords array returned HTTP 200 and 8,499 records for Fachinformatiker Systemintegration; the companion getNumberOfJobs endpoint reports 2.6 million live postings, and jv-search/search/getNumberOfJobs plus /jv/id/{id} give counts and detail. EURES job ids turn out to be base64 encodings of BA reference numbers, so the two sources deduplicate trivially. EURES is green-tier by construction: it is an official EU jobseeker service. Its search response embeds full text, which makes it useful for keyword-scoring without a second detail call.

EURES search - verified live, HTTP 200

```
curl -sS -X POST \
  "https://europa.eu/eures/api/jv-searchengine/public/jv-search/search" \
  -H "Content-Type: application/json" \
  -d '{"keywords":[{"keyword":"Fachinformatiker Systemintegration",
        "specificSearchCode":"EVERYWHERE"}],
      "locationCodes":["de"],"resultsPerPage":50,"page":1}'
```

4.3 Arbeitnow API

Arbeitnow is a German job board with a genuinely free public API and no key. The endpoint api/job-board-api returns 325 listings per page in a stable schema (slug, company_name, title, url, location, description, tags, job_types, remote, created_at) and updates hourly, ordered by created_at. Two caveats shape its use: the search parameter is ignored server-side, so the pipeline paginates one to three pages per run and filters client-side against the Chapter 2 keyword set; and the terms ask for a link back and no abuse, so a single-digit number of requests per hour is appropriate. During testing, an unfiltered page of 325 contained two Systemintegration-related listings, consistent with a board that skews toward software development but still carries infrastructure and support roles. Arbeitnow is the only no-key source besides the BA that is German-native, which earns it a permanent slot in the backbone.

Arbeitnow - verified live, HTTP 200

```
curl -sS "https://www.arbeitnow.com/api/job-board-api?page=1"
# hourly refresh, ordered by created_at; filter client-side
# against the FISI keyword list and the train-reachable town list
```

5. Tier 2 - Keyed Aggregator APIs

Aggregator APIs trade a one-time registration for server-side filtering that the no-key sources lack. Instead of pulling hundreds of listings and filtering locally, Asha's pipeline can push the keyword and location into the query and receive only relevant hits. Four aggregators matter for Germany; the rest of the aggregator landscape was checked and dismissed, and those results are preserved in Appendix B so that nobody re-tests them next month.

Provider	Registration	Endpoint shape	Free tier / limits
Adzuna	developer.adzuna.com (app_id + app_key)	GET api.adzuna.com/v1/api/jobs/de/search/{page}?app_id=..&app_key=..&what=systemintegration&where=N%C3%BCrnberg&max_days_old=7	25 req/min, 250/day, 1,000/week; attribution link required
Jooble	jooble.org/api (key generated after signup)	POST jooble.org/api/{key} with JSON {keywords, location, page}	Free tier quoted around 500 requests; simple POST semantics
Careerjet	careerjet.com/partners/api (v4)	Affiliate search API, locale de_DE; requires user_ip, user_agent, Referer	Free for affiliates; legacy keyless API permanently closed (verified 401)
WhatJobs	Publisher/affiliate program	GET api.whatjobs.com/api/v1/jobs.json with publisher key and user_ip	Unauthenticated calls verified rejected with HTTP 410

Table 5-1: Keyed aggregator APIs covering Germany. All four were live-verified during research.

Adzuna is the strongest of the four for this use case. Its German index is substantial, its query language supports what, where, max_days_old and category=it-jobs, and responses include created timestamps and salary fields, which feed the tracker's freshness scoring directly. The attribution requirement (a visible Jobs by Adzuna link when results are displayed) is trivially satisfied inside a personal CSV digest by adding a source column and a footer line. Jooble is the second recommendation: the POST interface is dead simple and its German coverage is core-market. Careerjet v4 and WhatJobs are optional extras; both work but add little beyond Adzuna plus Jooble for the effort of maintaining two more integrations. A fake-key probe against Adzuna returned a clean 401 with a pointer to its documentation, confirming the service is live and well-behaved.

Adzuna search - DE, keyword + geo + freshness

```
GET https://api.adzuna.com/v1/api/jobs/de/search/1
    ?app_id=YOUR_ID&app_key=YOUR_KEY
    &what=systemintegration&where=N%C3%BCrnberg
    &category=it-jobs&max_days_old=7

# Response fields used by the tracker:
#   title, company.display_name, location.display_name,
#   redirect_url, created, salary_min, salary_max, description
# Register at https://developer.adzuna.com (free)
```

6. Tier 3 - Portal-Native Channels

The big commercial portals offer no candidate-facing APIs: XING closed its public developer API years ago (only an employer-side e-recruiting API remains), StepStone's api.stepstone.com is partner-only, Indeed retired its Publisher API and switched job-search API access off in 2024, and LinkedIn's Jobs API is restricted to Talent Solutions partners. All historical RSS feeds are dead; Indeed's RSS was verified returning 403 on three domain variants, and StepStone answers even its robots.txt with an Akamai 403. Nevertheless two verified, login-free channels exist, and the portals' own alert products cover the rest legally and reliably.

6.1 XING server-rendered search results

XING's jobs search page renders server-side for logged-out visitors and embeds the full structured result set as JSON inside the HTML, in the serverData.APOLLO_STATE object. A plain curl that follows the 301 redirect from /jobs/search?keywords=... to the canonical /jobs/search/ki... URL returns HTTP 200 with roughly 16 jobs per page, each carrying title, slug URL, city, employment type, salary minimum, maximum and median in EUR, the apply URL with a flag for third-party redirects, a refreshedAt timestamp for new-detection, and an application type marker that identifies one-click-apply jobs. XING's GraphQL endpoint remains transport-live but only serves allow-listed persisted operations, so the SSR page is the practical extraction surface. Because XING's robots.txt disallows /jobs/search paths, this channel is yellow-tier: keep it to one or two requests per hour, personal use only, no redistribution, and back off immediately on any 403.

XING SSR extraction - verified live, HTTP 200, 363 KB page

```
curl -sSL "https://www.xing.com/jobs/search?keywords=Fachinformatiker
  &location=Erlangen" -H "User-Agent: Mozilla/5.0" \
  | grep -o "APOLLO_STATE.*" | head -c 200000

# Parse VisibleJob:* keys in APOLLO_STATE JSON.
# Fields per job: title, url/slug, city, employmentType,
#   salaryMin/salaryMax/salaryMedian (EUR), applyUrl,
#   redirectsToThirdPartyUrl, refreshedAt, one-click marker
```

6.2 LinkedIn guest endpoint

LinkedIn serves public job search results to logged-out visitors through the guest API at linkedin.com/jobs-guest/jobs/api/seeMoreJobPostings/search. The endpoint accepts keywords, location and start (paging in steps of 25) plus the freshness parameter f_TPR=r86400 for postings of the last 24 hours, and returns an HTML fragment of job cards with HTTP 200. It was verified live twice with Fachinformatiker and IT-Systemelektroniker queries against Erlangen, returning 30-31 KB fragments with 50 cards including real job URLs. LinkedIn throttles aggressively, so the tracker should stay under roughly 100 requests per day with jittered delays, never send account cookies, and treat this purely as a secondary freshness channel. The same ToS logic as XING applies: yellow tier, low volume, no

redistribution.

LinkedIn guest search - verified live, HTTP 200

```
curl -sS "https://www.linkedin.com/jobs-guest/jobs/api/
seeMoreJobPostings/search?keywords=Fachinformatiker
&location=Erlangen%2C+Bayern%2C+Germany&start=0
&f_TPR=r86400" -H "User-Agent: Mozilla/5.0"
# Parse job cards; page with start=25, 50, ... ; keep < 100 req/day
```

6.3 Native alerts for the defended portals

StepStone, Indeed, meinestadt.de, Jobware and Monster actively defend their search infrastructure (verified: Akamai connection resets on StepStone, DataDome on Monster, Cloudflare on glassdoor.de, Akamai 403s on all four meinestadt endpoints probed). The correct engineering answer is to use their own alert products, which are free, fast to set up, and legal by construction. Chapter 11 gives the exact configuration checklist; the essential trick for the CSV pipeline is an IMAP or Email-Parser step that converts alert emails into tracker rows, so even the daily-digest portals feed the same deduplicated list as the APIs. One further verification is worth recording: XING's one-click application (Bewerbung in einem Klick) is the fastest native apply flow of all German portals, requiring only a complete XING profile, and the SSR payload flags which postings support it, letting the tracker prioritize them.

7. Tier 4 - ATS-Direct Feeds and the Employer Watchlist

The deepest cut bypasses portals entirely: most German mid-size and large employers run their careers site on a commercial applicant tracking system (ATS), and most ATS products expose a public, undocumented or semi-documented feed of all openings. Watching ten relevant employers at the ATS layer often surfaces postings days before they reach aggregators, with zero competition from other applicants' tools. Every pattern in the table below was verified live during research, either at a generic demo tenant or, more importantly, at a real employer in Asha's commuting area.

ATS	Feed pattern	Format	Verified example (HTTP 200)
Personio	https://{co}.jobs.personio.de/xml	XML	nuernbergmesse-gmbh (56 positions), itonics-gmbh (15), nerdware (22)
SuccessFactors	https://{careers-host}/jobs/feed.xml or /feed.xml (follow redirects)	XML/RSS	jobs.schaeffler.com/feed.xml (5.6 MB, all jobs); careers.adidas-group.com/jobs/feed.xml (5.9 MB)
Workday (CXS)	POST https://{co}.wd{n}.myworkdayjobs.com/wday/cxs/{co}/{site}/jobs with JSON body	JSON	DATEV: datev.wd3.myworkdayjobs.com/wday/cxs/datev/Datev_Careers/jobs, searchText=Fachinformatiker
SmartRecruiters	https://api.smartrecruiters.com/v1/companies/{co}/postings	JSON	GfK absorbed into NielsenIQ: companies/NielsenIQ/postings?country=de
Greenhouse	https://boards-api.greenhouse.io/v1/boards/{co}/jobs	JSON	boards-api.greenhouse.io/v1/boards/contentful/jobs (pattern proof)
Lever	https://api.lever.co/v0/postings/{co}?mode=json (EU: jobs.eu.lever.co)	JSON	leverdemo tenant; real EU slugs mostly 404 in DE
Recruitee	https://{co}.recruitee.com/api/offers/	JSON	yourcareer.recruitee.com (pattern proof)
PUMA custom	https://about.puma.com/en/dd_job_search?search=&page=0	JSON	594 jobs in one response
Phenom	API tenant-locked; use sitemap fallback	XML sitemap	Siemens Healthineers: /global/en/sitemap_index.xml then sitemap1/2.xml, ~750 job URLs
In-house	https://career.{co}.com/sitemap.xml	XML sitemap	Hetzner: 142 URLs incl. live FISI-Ausbildung pages (j586/j588/j749/j751)

Table 7-1: ATS feed patterns with live verification evidence. {co} is the employer's tenant slug.

Workday CXS POST - DATEV (Nürnberg), verified live, HTTP 200

```
curl -sS -X POST \
  "https://datev.wd3.myworkdayjobs.com/wday/cxs/datev/
Datev_Careers/jobs" \
  -H "Content-Type: application/json" \
  -d '{"appliedFacets":{},"limit":20,"offset":0,
        "searchText":"Fachinformatiker"}'

# job detail: GET .../wday/cxs/datev/Datev_Careers/job/{externalPath}
```

Discovery recipes for new ATS tenants

- Google **site:jobs.personio.de {city}** to enumerate Personio tenants near a target city; the /xml suffix usually works on the same host.
- Career pages linking out with **utm_source=j2w** (Job2Web / SuccessFactors) almost always expose a /feed.xml one directory up.
- Any host matching **.wd{n}.myworkdayjobs.com** accepts the CXS POST shown above; the site path is in the careers URL.
- Phenom-powered careers (job pages under /job/R-xxxxx/) usually block the API but ship complete XML sitemaps: start at /sitemap.xml.

- Softgarden exposes wdteg.softgarden.io pages but no verified XML; treat as HTML-only for now.

Employer watchlist for the Erlangen-Nürnberg corridor

The watchlist below was assembled from live ATS verification plus IHK-directory and IT-cluster research. It is the concrete polling list for the ATS layer of the pipeline; each row notes the city and, where verified, the feed that Chapter 10's scheduler will call. The MSP and Systemhaus segment is listed by name because it is the classic first employer of FISI juniors, and such firms rarely have ATS feeds, so they are covered by the portal and aggregator layers instead of direct polling.

Employer	City	Segment / feed status
DATEV	Nürnberg	Software/IT services; Workday CXS verified
Siemens Healthineers	Erlangen/Forchheim	Medtech IT; Phenom sitemap verified
Siemens AG	Erlangen region	Avature; no public feed, use native alerts
Schaeffler	Herzogenaurach	Industry IT; SuccessFactors feed.xml verified
adidas	Herzogenaurach	SuccessFactors feed.xml verified
PUMA	Herzogenaurach	Custom JSON search verified
GfK / NielsenIQ	Nürnberg	Market research IT; SmartRecruiters verified
Hetzner	Nürnberg/Falkenstein	Hosting; sitemap verified, live FISI pages
N-ERGIE	Nürnberg	Utility IT; my-job-shop JSON needs runtime key, watch HTML
NürnbergMesse	Nürnberg	Personio /xml verified
ITONICS	Erlangen	Personio /xml verified
Nerdware	Nürnberg	Personio /xml verified
LEONI	Fürth	Industry IT; watch careers page
Nürnberger Versicherung	Nürnberg	Insurance IT; watch careers page
Brose	Bamberg	Industry IT; direct-train reachable, watch careers page
FAU (RRZE), Uniklinikum Erlangen, TH Nürnberg, Stadt Nürnberg IT, VAG, STAEDTLER, Faber-Castell, PLAYMOBIL (Zirndorf)	Erlangen/Nürnberg region	Public sector and mid-caps; steady FISI demand, watch career pages
noris network, netcup, OSB AG (Bamberg), ZOLLHOF startups	Nürnberg/Bamberg	Infrastructure/hosting/startups
Bechtle, CANCOM, Computacenter (branch offices)	Nürnberg/Fürth/Erlangen	Systemhaus majors; portal-layer coverage

Employer	City	Segment / feed status
ATA Solutions, IPExx, Lehmann IT, HWS Gruppe, BES Systemhaus, VIA APPIA, B2 Innovations	Mittelfranken	Regional MSPs; verify city and watch portals
Empfehlungsbund IT-Mittelstand members (it-mittelstand.de/nuernberg), BitMi directory entries	Nürnberg metro	Aggregated Mittelstand IT employers; periodic manual sweep

Table 7-2: 40-employer watchlist (rows with multiple names each expand to several employers). Feed statuses reflect live verification.

8. Remote and Niche RSS Suite

Remote-first boards round out the all-Germany fallback: a fully remote junior support or sysadmin role satisfies the train constraint trivially because the commute is empty. All four services below expose key-less feeds, were verified live, and impose only attribution-style conditions. WeWorkRemotely publishes per-category RSS; the DevOps/sysadmin feed is the FISI-relevant one (its older it-and-support slug now 301-redirects). Himalayas offers one clean RSS endpoint. Remotive's JSON API is free with mandatory attribution and a 24-hour publication delay, which matters only because the tracker must not treat Remotive as a first-alert channel. RemoteOK serves a JSON array whose first element is a legal notice; its tags include junior and sysadmin, useful for filtering, while its RSS endpoint is retired (410).

Source	Endpoint	Format	Notes
WeWorkRemotely	weworkremotely.com/categories/remote-devops-sysadmin-jobs.rss (plus programming, customer-support, product feeds)	RSS	25 items per feed; pubDate freshness; all verified 200
Himalayas	himalayas.app/jobs/rss	RSS	Single endpoint; simple schema; verified 200
Remotive	remotive.com/api/remote-jobs?limit=50	JSON	Attribution mandatory; 24 h delay on free tier; verified 200
RemoteOK	remoteok.com/api	JSON	First array element is a legal notice; epoch + tags fields; verified 200

Table 8-1: Remote feed suite. All endpoints verified live during research.

Remote suite - one-line health checks, all verified 200

```
curl -sS "https://weworkremotely.com/categories/
  remote-devops-sysadmin-jobs.rss" | head -c 300
curl -sS "https://himalayas.app/jobs/rss" | head -c 300
curl -sS "https://remotive.com/api/remote-jobs?limit=5" | head -c 300
curl -sS "https://remoteok.com/api" | head -c 300
```

9. The Erlangen Train-Reachability Filter

Asha's constraint is precise: jobs within roughly 50 km of Erlangen that are directly reachable by train, without changing. Distance filters are the wrong tool, because rail geography is nonlinear; Coburg at about 65 km is a 47-minute direct ride, while rail-less Herzogenaurach at 15 km is unreachable by Asha's definition. The correct instrument is a transit-data filter built on the VGN (Verkehrsverbund Großraum Nürnberg) GTFS feed, published as open data under CC BY-SA 3.0 DE at vgn.de/opendata. The feed contains 25,464 stops and 90,753 trips, and its stop IDs embed the official AGS municipality key (de:09562:* = Erlangen, de:09563:* = Fürth, and so on), which allows direct mapping from stations to municipalities without any geo-join.

Source	URL	License / status
VGN GTFS (primary)	vgn.de/opendata - stable GTFS.zip link	CC BY-SA 3.0 DE; updated irregularly (last: June 2026); verified download
gtfs.de Germany feeds	download.gtfs.de/germany/{nv,rv,fv}_free/latest.zip	Open; daily refresh; verified; fallback when VGN is stale
v6.db.transport.rest	community DB API proxy	Verified dead for data (503 on all endpoints since ~2025); do not build on it
db-vendo (GitHub)	public-transport/db-vendo-client	Maintained library but no hosted instance; upstream DB APIs unreliable; self-host only

Table 9-1: Transit data sources for the reachability filter, with live verification status.

The computation is a zero-transfer expansion: load the GTFS feed, collect all platforms of Erlangen Bahnhof (parent stop de:09562:3110), keep every trip of route_type 2 (rail) that touches any of them, and take the union of the other stops on those same trips, which are reachable with zero transfers by construction. In-vehicle minutes are arrival minus Erlangen departure, and services are filtered to those active on at least 60 percent of weekdays in a 90-day window to kill holiday ghosts. Running this procedure on the June 2026 feed yields 82 stops across 29 municipalities within 60 minutes, the whitelist below. The pragmatic pipeline uses this table as a hardcoded filter, refreshed quarterly after each GTFS release, because journey-API approaches (v6.db.transport.rest) are verified dead.

Municipality	Station(s)	Line	~min	Direction
Erlangen (all districts)	Hbf, Paul-Gossen-Str., Bruck, Eltersdorf	S1	2-6	origin
Fürth	Hbf, Klinikum, Vach	RE/S1	7-17	south

Municipality	Station(s)	Line	~min	Direction
Nürnberg	Hbf + S1 city stops (Dürrenhof ... Laufamholz)	all	14-41	south
Baiersdorf	Baiersdorf	S1	6	north
Bubenreuth	Bubenreuth	S1	3	north
Forchheim	Forchheim, Forchheim Nord, Kersbach	RE/S1	7-18	north
Eggolsheim	Eggolsheim	RE14/S1	16	north
Buttenheim	Buttenheim	S1	24	north
Hirschaid	Hirschaid	RE/S1	16	north
Strullendorf	Strullendorf	S1	31	north
Bamberg	Bamberg	RE/IC61/S1	20-37	north
Hallstadt	Hallstadt	RE14/28	30	north
Breitengüßbach	Breitengüßbach	RE14/28	30	north
Rattelsdorf	Ebing	RE14/28	34	north
Zapfendorf	Zapfendorf	RE14/28	36	north
Ebensfeld	Ebensfeld	RE14/28	38	north
Bad Staffelstein	Bad Staffelstein	RE14/28	42	north
Lichtenfels	Lichtenfels, Schney	RE14/28/IC61	37-57	north
Michelau i.OFr.	Michelau	RE14	53	north
Hochstadt a.Main	Hochstadt-Marktzeuln	RE14	56	north
Redwitz a.d.Rodach	Redwitz	RE14	59	north
Kronach	Kronach	IC61 (limited), RE14 65	51 flag	north
Haßfurt	Haßfurt	RE20	41	west
Schwaig b.Nbg	Schwaig	S1	43	east
Röthenbach a.d.Pegnitz	Röthenbach, Steinberg, Seespitze	S1	46-50	east
Lauf a.d.Pegnitz	Lauf West, Lauf	S1	52-53	east
Ottensoos	Ottensoos	S1	57	east
Henfenfeld	Henfenfeld	S1	60	east
Coburg	Coburg	RE19/28/29	47	north (65 km)
Ansbach	Ansbach	IC61 only, about 4/day	48 flag	south

Table 9-2: Verified zero-transfer whitelist from Erlangen Bahnhof, 60 minutes or less (VGN GTFS, June 2026). Flagged rows are limited-frequency and need a schedule check per application.

Exclusions are as decision-relevant as inclusions. Herzogenaurach hosts PUMA and adidas yet has no rail at all, bus-only, so it fails the constraint despite the strongest employer names; Höchstadt a.d.Aisch is likewise rail-less. Schwabach, Roth, Feucht, Altdorf and Neumarkt sit one transfer away on S2/S3 lines from Nürnberg, and the S1's brief extension toward Neumarkt was reverted on 14 December 2025, so stale web sources claiming otherwise should be ignored. Würzburg, Kitzingen and Neustadt an der Aisch are bypassed entirely by the rail lines that serve Erlangen; Gräfenberg and Eckental require a transfer; and anything past 60 minutes (Rödental 70, Hersbruck 63, Hartmannshof 74) is out. For filtering implementation the pipeline matches postings against this whitelist by town name and PLZ prefix, with the flagged entries (Kronach, Ansbach) kept behind a warning rather than dropped, since a strong FISI offer with four daily IC61 connections may still be worth an interview conversation.

Tariff context completes the picture: Nürnberg and Fürth sit in zones 100/200 (Stadttarif A), Erlangen in zone 400, and the Deutschlandticket at 63 EUR per month from January 2026 covers the entire whitelist at zero marginal per-ride cost. Economically, then, the filter should be time-based rather than distance-based, exactly as the GTFS computation above already is.

10. The CSV Tracker Pipeline

All components assemble into one pipeline whose output is exactly what Asha asked for: a deduplicated CSV tracker of fresh, train-reachable, FISI-relevant postings, refreshed automatically and scored so the best application targets are always at the top. The architecture is deliberately boring and debuggable: Python 3 standard library plus requests, flat files, one cron entry per source group, and no database beyond a seen-jobs store.

Pipeline architecture (data flow)

```
cron (every 30-60 min)
   |
   +-- fetch_ba.py        BA API v6 keyword loop  (green)
   +-- fetch_eures.py     EURES POST search       (green)
   +-- fetch_arbeitnow.py page 1-3 + client filter(green)
   +-- fetch_adzuna.py    what/where/max_days_old (green, keyed)
   +-- fetch_ats.py       DATEV/Schaeffler/adidas/PUMA/GfK
   |                      /Personio trio/Hetzner  (green)
   +-- fetch_rss.py       WWR + Himalayas         (green)
   +-- fetch_portals.py   XING SSR + LI guest     (yellow, 1-2/h)
   |
   v
normalize -> dedupe (BA refnr | URL | title+company+plz hash)
   -> train-filter (whitelist Table 9-2)
   -> score (freshness + keywords + salary flag + one-click)
   -> jobs.csv (sorted) + seen.json (state)

optional: LLM scorer re-ranks top N against Asha's CV text
```

The polling schedule respects each source's tolerance. The BA keyword loop (19 German core and variant terms) runs hourly at one request per keyword, which is 19 requests spread over a minute - far inside any reasonable reading of intended use. Arbeitnow and the RSS suite poll hourly. Adzuna polls every two hours for fresh postings only (max_days_old=3), staying far below its 250-call daily budget. The ATS group polls every two to four hours depending on feed size; the two megabyte SuccessFactors feeds should be fetched with streaming and a conditional ETag if available. The yellow-tier portal channels run at most twice per hour, never in parallel, with jitter. Alert emails from StepStone, Indeed and meinestadt arrive daily and are ingested by a small IMAP parser so that every source lands in the same CSV.

Source group	Interval	Requests/run	Daily budget
BA API keyword loop (19 queries)	60 min	19 + details	about 480, green-tier intended use
EURES search	60 min	1-2	uncapped documented, keep low
Arbeitnow pages 1-3	60 min	3	72, trivial
Adzuna DE fresh query	120 min	2	24 of 250/day
ATS feeds (10 employers)	120-240 min	10	about 80, all green
Remote RSS suite	60 min	6	144, trivial
XING SSR + LinkedIn guest	30-60 min each	2-3	under 100 total, yellow-tier discipline

Table 10-1: Polling schedule and request budgets.

Scoring model

Every new row gets a 0-100 score from four transparent components, so Asha can tune weights without touching code internals: freshness (within 24 h starts at 40 points, decaying daily), keyword match density from the Chapter 2 term lists as matched against title plus description, location bonus from the train whitelist (zero-transfer within 40 minutes scores highest; flagged towns score mid), and friction bonuses for one-click apply support, salary transparency and homeoffice flags. Rows below a threshold of 30 are written to the CSV but marked skipped, keeping the file a complete audit trail while the daily review reads only the top segment. An optional LLM pass re-ranks the top twenty rows against Asha's CV text with a strict-JSON prompt (score plus one-line reason), which costs fractions of a cent per run and catches semantic matches that keyword counting misses, such as Client-Management postings titled as IT-Service-Techniker.

Core fetcher skeleton (Python, requests only)

```
import requests, time
BASE = "https://rest.arbeitsagentur.de/jobboerse/jobsuche-service"
KEY  = {"X-API-Key": "jobboerse-jobsuche"}
KW   = ["Fachinformatiker Systemintegration", "Systemintegration",
        "IT-Systemelektroniker", "Systembetreuer", "IT-Administrator"]

def ba_search(kw, page=1):
    r = requests.get(f"{BASE}/pc/v6/jobs", headers=KEY, params={
        "was": kw, "wo": "Erlangen", "umkreis": 50,
        "page": page, "size": 100,
        "veroeffentlichtseit": 7, "angebotsart": 1}, timeout=30)
    r.raise_for_status()
    return r.json()["stellenangebote"]  # see field table, App. C

for kw in KW:
    rows = ba_search(kw)
    time.sleep(3)   # polite pacing
    # -> normalize into the shared row schema, then dedupe
```

CSV column	Content	Purpose
first_seen	ISO timestamp of pipeline detection	Speed KPI; enforce the 72 h rule
published	Source publication date	Freshness score
title	Posting title	Review
company	Employer name	Review and dedupe
town	Location as posted	Whitelist match
train	Direct / flagged / no	Commute decision
source	BA / EURES / Arbeitnow / Adzuna / ATS:{co} / XING / LI / RSS:{name} / mail:{portal}	Traceability and attribution
url	Application URL	One click to apply
salary	Range when exposed (BA, XING, Adzuna)	Prioritization
score	0-100 pipeline score	Sort key
status	new / skipped / applied / rejected / interview	Manual tracking column Asha edits

Table 10-2: The tracker schema. Status is the only column updated by hand.

Cron wiring is three lines: a morning full run at 06:40 that also rebuilds the CSV header and a midday increment at 12:40, plus a weekly Sunday job that re-validates all source health by replaying the verification curls from this report and emails a diff summary. The seen.json state file keyed by the dedupe hash guarantees that review effort only ever goes to genuinely new rows. For Asha this means one five-minute morning ritual: open the CSV sorted by score, apply to everything above 70 with the native one-click flows, mark statuses, done.

11. Portal-Native Speed Tactics

Programmatic detection only wins applications if the apply path is fast too. This chapter is the manual half of the system: how to configure each portal's own alerting for fastest delivery, and how to keep every application to a one-to-two minute action. Everything here is free and officially supported, which makes it the safety net that keeps coverage complete even where scraping is off the table.

Portal	Alert product	Fastest delivery	Setup notes
XING	Job-Alerts (up to 4+ searches)	Prompt/push fastest, email second	Create one alert per keyword cluster; complete profile to unlock Bewerbung in einem Klick
StepStone	Jobagent	Daily digest email	Several narrow agents beat one broad one; Schnellbewerbung needs an uploaded CV since 2022
Indeed DE	Email alerts + app push	Push faster than email	Alert on multiple queries; profile enables Indeed Apply
LinkedIn	Job alerts	Push near-instant	Set alerts without Easy-Apply automation; guest endpoint covers detection separately
meinestadt	Jobagent email	Daily	Covers a regional audience the big portals miss
Jobware	Jobagent email	Daily	Legacy but still relevant for Bavarian Mittelstand
BA Jobbörse	Suchagent email	Daily	Redundant with the API but useful as an integrity check of the pipeline

Table 11-1: Native alert matrix. Daily digests are the fallback layer; the pipeline covers real-time detection.

- Complete every portal profile once, properly: XING and StepStone one-click flows pull straight from the profile, and incomplete profiles silently disqualify the fast path.
- Upload a clean PDF CV everywhere; ATS-parsed profiles raise match scores on StepStone and Indeed.
- Keep two ready-to-paste Anschreiben skeletons: one for MSP/Systemhaus roles emphasizing customer-facing service and Windows/network basics, one for corporate IT emphasizing structure and tickets.
- Apply inside 72 hours of publication; the pipeline's first_seen column makes this measurable, and Chapter 2's evidence says the interview payoff is up to 8x.
- Prefer direct employer URLs (ATS feeds) over portal redirects when both exist; direct applications showed an 11.2 percent interview rate in aggregate tracking data versus lower rates through intermediaries.
- Log every application in the tracker's status column the same day; the response-time statistics from Chapter 2 only become actionable once Asha measures his own funnel.

12. Legal and Etiquette

German law on collecting job postings is favorable to a personal pipeline as long as it stays personal. Job advertisements are business data, not personal data, so GDPR is largely beside the point; the real constraints are the database-maker's right of §87b UrhG, contract terms (AGB), and unfair-competition law (UWG). The governing case line is consistent: the BGH's Flugbuchungssystem II decision (I ZR 57/17, 30 January 2019) holds that scraping is not unfair per se, but terms of service that ban automated access are valid contract clauses whose breach supports injunctions; the CJEU's Ryanair v PR Aviation (C-30/14) confirms ToS bans remain enforceable even where database protection does not apply; and Innoweb v Wegener (C-202/12) treats repeated small extractions as a substantial part of a database. The practical translation: green-tier sources are unproblematic, yellow-tier channels are a contract risk profiled at the IP-ban level for low-volume personal use, and red-tier scraping is where Abmahnung exposure begins.

Source / method	Risk	Reasoning and mitigation
BA Jobsuche API, EURES API	green	Official open offers for jobseekers; intended personal use; poll politely, cache, never re-host the data
Arbeitnow API	green	Explicitly free public API; link back and do not abuse
Adzuna / Jooble / Careerjet / WhatJobs	green	Registered keys, documented limits; attribution required on Adzuna and Careerjet
ATS feeds (Personio /xml, feed.xml, CXS, SmartRecruiters)	green	Feeds are published for consumption; poll at humane intervals, keep source links
RSS suite (WWR, Himalayas, Remotive, RemoteOK)	green	Feed publication implies consumption permission; 30-60 min minimum interval, attribution per terms
XING SSR extraction (logged out)	yellow	robots.txt disallows /jobs/search; ToS forbid automation; low-volume personal polling is the practical risk ceiling (IP ban); 1-2 requests/hour, back off on 403, never redistribute
LinkedIn guest endpoint	yellow	Same ToS situation plus aggressive throttling; no login cookies ever; well under 100 requests/day
StepStone / Indeed / meinestadt scraping	red	Active anti-bot (Akamai, DataDome, Cloudflare) plus explicit ToS bans plus robots 403; use native alerts instead
Circumventing Cloudflare/Akamai	red	Wilful breach; potential §303a StGB exposure at scale; excluded from this design by construction

Table 12-1: Legal risk matrix. The pipeline as designed touches only green and, optionally, tiny volumes of yellow.

- Store links and metadata, not full ad texts; the tracker schema in Table 10-2 already does this and keeps the dataset clearly below any substantial-part threshold.
- Keep personal data out of the CSV: recruiter names or contact persons in ads are stored minimally or not at all (recruiter personal data is personal data).

- Never republish the collected data; the CSV is a personal work instrument, not an input for a public board.
- Respect robots.txt and visible 403/429 responses as hard stop signals; etiquette is also the cheapest anti-lawyer strategy.
- The pipeline needs zero red methods; if a source turns hostile, drop it and widen the green tier, which Chapter 3 shows already covers the market adequately.

13. The 14-Day Action Plan

Everything converges into a two-week launch plan that ends with a running pipeline, a fully tuned application machine, and the first wave of fast applications out the door. The plan assumes evenings and one weekend, no budget beyond a Deutschlandticket and about 100 EUR of certification material, and produces measurable outputs each day so momentum stays visible.

Day	Actions	Output
1	Set up XING, StepStone, Indeed, LinkedIn profiles completely; upload PDF CV; register Adzuna and Jooble keys	Profiles live; API keys in .env
2	Create native alerts per Table 11-1; set up the tracker folder, Python venv, cron skeleton	Alert emails flowing; repo ready
3	Implement fetch_ba.py (Chapter 10 skeleton) with the full 19-keyword loop; verify counts against Table 4-1	Backbone rows in jobs.csv
4	Add fetch_eures.py, fetch_arbeitnow.py, fetch_rss.py; implement dedupe and seen.json	Four green sources merged
5	Add fetch_adzuna.py and Jooble POST; add source columns and attribution footer	Aggregator layer live
6	Implement fetch_ats.py for the Table 7-1 verified feeds (DATEV, Schaeffler, adidas, PUMA, GfK, Personio trio, Hetzner)	ATS layer live
7	Implement the train-reachability filter (Table 9-2 as data) and the scorer; first full CSV build	Ranked tracker output
8	Add the yellow-tier portal pollers with rate limits and jitter; add IMAP ingestion of alert emails	All sources in one CSV
9	Morning ritual begins: apply to all rows above score 70 within the 72 h window	First fast applications out
10	Write the two Anschreiben skeletons; tailor one per top-10 employer segment	Apply time under 2 minutes
11	Start AZ-900 study block 1; schedule the exam	Certification in motion
12	Direct outreach wave: five short emails to MSP watchlist employers referencing their current postings	Warm contacts created
13	AZ-900 study block 2; review tracker KPIs (rows/day, applications, first_seen-to-applied delta)	Funnel metrics baseline
14	Weekly review ritual: source health replay (verification curls), keyword tuning, drop dead sources, promote new whitelisted towns	Self-improving pipeline

Table 13-1: The 14-day launch plan. After day 14 the system runs on the weekly review loop.

From week three onward the routine is five minutes each morning plus a 30-minute Sunday review, with the pipeline doing everything else. The KPI set is small and honest: new relevant rows per day, median hours from publication to first_seen, median hours from first_seen to applied, and response rate after two weeks. If the first_seen-to-applied median exceeds 72 hours, the morning ritual is failing and needs shortening; if relevant rows per day drops below one, the keyword set or the whitelist needs widening. The Deutschlandticket turns the flagged towns of Table 9-2 into realistic commutes, and by the time the first interviews arrive, AZ-900 on the CV removes the last excuse a screening HR filter had. That is the whole system: see everything first, apply fastest, measure, repeat.

Appendix A. Master URL Directory

All operational URLs in one place, grouped by function, so the verification replay of day 14 can be scripted directly from this table.

Group	URL	Purpose
BA API docs	https://jobsuche.api.bund.dev/	Official OpenAPI spec; documents the public key
BA API search	https://rest.arbeitsagentur.de/jobboerse/jobsuche-service/pc/v6/jobs	Primary search endpoint
BA API details	https://rest.arbeitsagentur.de/jobboerse/jobsuche-service/pc/v4/jobdetails/{base64(refnr)}	Full posting text
EURES	https://europa.eu/eures/api/jv-searchengine/public/jv-search/search	EU search, no auth
Arbeitnow	https://www.arbeitnow.com/api/job-board-api	Free German board API
Adzuna	https://developer.adzuna.com/ + https://api.adzuna.com/v1/api/jobs/de/search/{page}	Registration and endpoint
Jooble	https://jooble.org/api	Key generation and docs
Careerjet v4	https://www.careerjet.com/partners/api	Partner signup
XING search	https://www.xing.com/jobs/search?keywords={kw}&location={city}	SSR JSON extraction (yellow)
LinkedIn guest	https://www.linkedin.com/jobs-guest/jobs/api/seeMoreJobPostings/search	Guest search (yellow)
Personio	https://{co}.jobs.personio.de/xml	ATS XML feed pattern
SuccessFactors	https://{careers-host}/jobs/feed.xml	ATS XML feed pattern
Workday CXS	https://{co}.wd3.myworkdayjobs.com/wday/cxs/{co}/{site}/jobs	ATS JSON POST pattern
SmartRecruiters	https://api.smartrecruiters.com/v1/companies/{co}/postings	ATS JSON pattern
Greenhouse	https://boards-api.greenhouse.io/v1/boards/{co}/jobs	ATS JSON pattern
Lever	https://api.lever.co/v0/postings/{co}?mode=json	ATS JSON pattern
Recruitee	https://{co}.recruitee.com/api/offers/	ATS JSON pattern
DATEV feed	https://datev.wd3.myworkdayjobs.com/wday/cxs/datev/Datev_Careers/jobs	Verified CXS POST
Schaeffler feed	https://jobs.schaeffler.com/feed.xml	Verified SF XML
adidas feed	https://careers.adidas-group.com/jobs/feed.xml	Verified SF XML
GfK/NIQ feed	https://api.smartrecruiters.com/v1/companies/NielsenIQ/postings	Verified SR JSON
PUMA feed	https://about.puma.com/en/dd_job_search?search=&page=0	Verified custom JSON

Group	URL	Purpose
Healthineers sitemap	https://jobs.siemens-healthineers.com/global/en/sitemap_index.xml	Verified sitemap fallback
Hetzner sitemap	https://career.hetzner.com/sitemap.xml	Verified sitemap
WWR RSS	https://weworkremotely.com/categories/remote-devops-sysadmin-jobs.rss	Verified remote feed
Himalayas RSS	https://himalayas.app/jobs/rss	Verified remote feed
Remotive API	https://remotive.com/api/remote-jobs	Verified remote JSON
RemoteOK API	https://remoteok.com/api	Verified remote JSON
VGN GTFS	https://www.vgn.de/opendata/	Transit open data for the train filter
gtfs.de	https://download.gtfs.de/germany/rv_free/latest.zip	Fallback regional GTFS

Table A-1: Operational URL directory. {co} denotes a tenant slug to substitute.

Appendix B. Dead-End Register

These channels were investigated and verified dead, blocked, or restricted during the research window. The register exists so that no future maintenance session re-tests them on a hunch; each row states the observed failure mode. If one of these revives, it graduates back into the coverage matrix of Chapter 3.

Channel	Observed status	Evidence
BA old API /jobads/jobs + jobsearch-beta	Dead (403)	All header and key variations returned 403 on 2026-10-12
Indeed RSS (rss.indeed.com, de.indeed.com/rss)	Dead (403)	Three domain variants verified 403
Indeed job search API	Off since 2024	Publisher program retired; page returns login wall (401)
StepStone scraping	Blocked	Akamai connection reset and robots.txt 403
Monster.de	Blocked + shrinking	DataDome challenge; brand sold to Bold Holdings after the 2025 bankruptcy
glassdoor.de	Blocked	Cloudflare challenge; merged into Indeed July 2026
meinestadt.de endpoints	Blocked	Akamai 403 on page, /jobs/rss and /rss
Jobware	Blocked	403 and connection kill; same platform family as kalaydo
GermanTechJobs /api/jobs	Redirects to paid service	302 to devitjobs.jobcopilot.com/signup
Jobicy gf-feed	404	All parameter variants rejected
Joblift	Defunct	CloudFront error domain-wide

Channel	Observed status	Evidence
jobs.munich-startup.de	Defunct	DNS NXDOMAIN
kimeta, JOBworld RSS, jobs.de, jobtensor /API, Get-in-IT jobboerse, heise/t3n/Golem jobs RSS	Bot walls / 404 / 503	No individual-access API verified
v6.db.transport.rest	Dead for data	503 on all endpoints since about January 2025
Careerjet legacy public API	Closed to new users	401 pointing to the v4 partner API
WhatJobs unauthenticated	Rejected	410 Unknown publisher
Talent.com API	Contract-only	CPC partner program, not individual-realistic
Remotive RSS-style instant data	24 h delay on free tier	Paid API exists at 5,000 USD/month; irrelevant here

Table B-1: Verified dead ends. Do not re-test without cause.

Appendix C. BA API Field Reference

Ground-truth field reference for the v6 search response, compiled from a live 100-item sample. The published OpenAPI schema is partially outdated, so this table reflects what the service actually returns today. Fields marked absent no longer exist despite lingering documentation on the internet.

Field	Type	Meaning
referenznummer	string	Unique posting key; also the detail-endpoint input after base64 encoding
stellenangebotsTitel	string	Ad title
hauptberuf / alleBerufe	string / array	Primary and all occupational classifications
stellenangebotsart	enum	ARBEIT, AUSBILDUNG, PRAKTIKUM; use angebotsart=1 to filter
firma	string	Employer name
stellenlokationen[]	array	One or more locations with adresse.ort, .plz, .region, .land plus breite/laenge coordinates
entfernung	number	Distance in km from the wo search center
datumErsteVeroeffentlichung	date	First publication date; the freshness anchor
aenderungsdatum	datetime	Last modification
eintrittszeitraum	object	Start window (von/bis)

Field	Type	Meaning
gehaltsspanneVon / Bis	number	Salary range when disclosed (artDerVerguetung=GEHALTSSPANNE)
verguetungsangabe	enum	Compensation disclosure type
arbeitszeitVollzeit / Teilzeit / ...	booleans	Working-time flags (vz, tz, snw, ho, mj)
homeofficemoeglich	boolean	Remote work possible
quereinstiegGeeignet	boolean	Career-changer suitable; useful junior signal
vertragsdauer	enum	Contract duration (UNBEFRISTET etc.)
arbeitgeberKundennummerHash	string	Hashed employer id; joins to the logo service
externeURL	string	External application URL; presence marks private-board referrals
hashId / externBoersenId	absent	Removed with the old API; do not parse for them

Table C-1: v6 search response fields as observed live (October 2026).

The detail endpoint returns the full stellenangebotsBeschreibung plus contact and application procedures, and the employer logo service under /vermittlung/ag-darstellung-service/ct/v1/arbeitgeberlogo/{hash} serves webp images, returning 404 simply when no logo exists. Rate behavior observed: no documented numeric limits, 12 rapid requests all 200, and an X-API-BLOCKED response header present on every call, which is the signal to watch for if the BA ever tightens access. Keep the pipeline's pacing at the levels of Table 10-1 and this API will keep working as the free backbone it is designed to be.